Report

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1. ROLE ENFORCEMENT

The ERP uses TWO databases:

• authdb → stores login credentials (users_auth table)

• erpdb → stores academic data (courses, grades, registrations, etc.)

The "users_auth" table contains:

- user_id
- username
- password_hash
- role (ADMIN, STUDENT, INSTRUCTOR)

When a user logs in through LoginFrame.java:

1. The system queries users_auth:

```
SELECT user_id, role, password_hash FROM users_auth WHERE username = ?
```

2. Password is checked using BCrypt.
3. The role determines which dashboard is launched:

• role = ADMIN → AdminDashboard

• role = STUDENT → StudentDashboard

• role = INSTRUCTOR → InstructorDashboard (if implemented)

All key operations are protected through ROLE CHECKS.

Examples:

• AdminUsersPanel → only Admin can add/edit/delete users

• AdminCoursesPanel → only Admin can manage courses

• Students cannot access course editing or user management

• Students can only register for courses assigned in timetable

This guarantees that no unauthorized user can perform administrative actions.

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2. MAINTENANCE MODE ENFORCEMENT

Maintenance mode is stored centrally inside the database table:

erpdb.settings

Structure:

```
--------------------------------------
setting_key    | setting_value
--------------------------------------
maintenance    | ON / OFF
--------------------------------------
```

Whenever the Admin toggles the switch:

- ToggleSwitch sends new state to the DB:

```
UPDATE settings SET setting_value='ON' WHERE setting_key='maintenance';
UPDATE settings SET setting_value='OFF' WHERE setting_key='maintenance';
```

- The yellow footerBanner becomes visible in AdminDashboard.

StudentDashboard, when opening any panel, checks:

SELECT setting_value FROM settings WHERE setting_key='maintenance';

If the value is "ON":

• Registration buttons become disabled

• Drop buttons become disabled

• Student sees a yellow banner

• Students can VIEW courses but cannot MODIFY any data

This ensures the entire system becomes read-only for non-admin users.

Only the ADMIN can turn maintenance mode OFF again.

Table Lists: